

A Course End Project Report on
SECURE PASSWORD MANAGEMENT SYSTEM

A8612 – Information Security

Submitted by

K. Bhanu Dheeraj

23881A0528

Course Facilitator

Ms. Dr. VIYYAPU LOKESHWARI VINYA
ASSOCIATE PROFESSOR, CSE

Department of COMPUTER SCIENCE AND ENGINEERING

Vardhaman College of Engineering, Hyderabad

March 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **“Secure Password Management System”** is carried out by **“K. Bhanu Dheeraj”** with Roll Numbers **“23881A0528”** towards **A8612 – Information Security** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **COMPUTER SCIENCE AND ENGINEERING** during the Academic year 2025-26.

Signature of the Course Faculty
Dr. VIYYAPU LOKESHWARI VINYA
ASSOCIATE PROFESSOR
CSE

Signature of the HOD
Dr. RAMESH KARNATI
ASSOCIATE PROFESSOR AND
HEAD, CSE

Contents

1	Abstract	2
2	Introduction	3
3	Problem Statement	4
4	Literature Survey	5
5	Methodology.....	6
6	System Description.....	7
7	Results.....	8
8	Conclusion.....	12
9	Future Scope.....	13
10	Reference.....	14

1.Abstract

With the increasing number of online services, secure password management has become a critical cybersecurity concern. This study presents the design and implementation of a Secure Password Management System that ensures safe storage and retrieval of user credentials. The system utilizes AES-256 encryption for securing stored passwords and bcrypt hashing for protecting user authentication credentials. A JWT-based authentication mechanism is implemented to ensure secure session management, while MongoDB is used for structured and isolated data storage per user. The proposed system demonstrates secure encryption-decryption workflows, user-based data separation, and protection against unauthorized access. The significance of this work lies in providing a practical, scalable, and secure password management solution aligned with modern cybersecurity standards.

2 Introduction

2.1 Background of the Domain

In the digital era, individuals and organizations rely heavily on online platforms for communication, banking, education, e-commerce, and cloud services. This rapid digital transformation has significantly increased the number of accounts users must manage. As a result, password management has become a fundamental aspect of cybersecurity. Traditional methods such as writing passwords in notebooks, storing them in plain text files, or reusing the same password across multiple platforms pose serious security risks. Modern cybersecurity practices emphasize encryption, secure authentication mechanisms, and proper data storage techniques to protect sensitive information from breaches and unauthorized access.

2.2 Importance of the Problem

Password-related vulnerabilities remain one of the leading causes of data breaches worldwide. Weak passwords, password reuse, phishing attacks, and improper storage mechanisms expose users to identity theft and financial loss. Many users struggle to create and remember strong, unique passwords for each service. Without a secure password management solution, sensitive credentials can be easily compromised. Therefore, implementing a secure system that encrypts stored passwords, ensures user authentication, and prevents unauthorized access is essential for maintaining data confidentiality and integrity.

2.3 Need for the Study

Despite the availability of commercial password managers, there is a growing need to understand and develop secure password storage systems using modern encryption standards. Studying the architecture and implementation of a secure password management system provides insight into real-world cybersecurity practices such as AES encryption, password hashing (bcrypt), JWT-based authentication, and secure database management. This study addresses the need for a practical and scalable solution that ensures user data isolation, encrypted storage, and secure access control.

2.4 Objectives of the Case Study

The primary objectives of this case study are:

1. To design and implement a secure password management system using modern cryptographic standards.
2. To apply AES-256 encryption for protecting stored credentials.
3. To use bcrypt hashing for secure user authentication and master password protection.

3 Problem Statement

In the modern digital ecosystem, individuals and organizations rely heavily on online platforms for communication, financial transactions, education, and cloud-based services. As the number of digital accounts increases, users are required to create and manage multiple passwords. This often results in insecure practices such as password reuse, weak password creation, or storing credentials in plain text formats. These behaviors significantly increase vulnerability to cyber threats including phishing attacks, brute-force attacks, credential stuffing, and data breaches. The absence of a secure, encrypted, and user-isolated password management system poses a serious risk to data confidentiality and personal privacy.

This problem affects a wide range of users, including individual internet users, students, working professionals, small businesses, and organizations that depend on secure authentication mechanisms. Individuals managing multiple online accounts are particularly vulnerable, as they struggle to remember complex passwords and often resort to unsafe storage methods. Organizations are also impacted when weak password practices compromise internal systems and sensitive information.

3.1 Scope and Boundaries of the Study

The scope of this study is to design and implement a secure web-based password management system that ensures encrypted storage and controlled access to user credentials. The system utilizes AES-256 encryption for protecting stored passwords, bcrypt hashing for securing login and master passwords, and JWT-based authentication for secure session management. Additionally, it ensures user-specific data isolation within a MongoDB database. However, the system is limited to a web-based implementation and does not include advanced features such as multi-factor authentication or enterprise-level deployment capabilities. The focus remains on building a secure, scalable, and practical solution aligned with modern cybersecurity standards.

4 Literature Review

4.1 Previous research

Many studies and commercial systems focus on secure password management using encryption and hashing techniques. Popular password managers use AES encryption to protect stored passwords and bcrypt or similar algorithms to secure master passwords. Modern web applications also use JWT-based authentication to maintain secure user sessions. These systems show that combining encryption, hashing, and authentication improves overall security.

4.2 Limitations

Most commercial password managers are paid and closed-source, which limits transparency. Some research focuses only on theoretical concepts without building a complete working system. Certain implementations also lack proper user data separation or secure integration of encryption and authentication mechanisms.

4.3 Why Needed

This case study is necessary to bridge the gap between theoretical research and practical implementation of secure password management systems. It provides a complete, web-based solution that integrates AES-256 encryption for vault security, bcrypt hashing for user and master password protection, JWT-based authentication for session management, and MongoDB for structured user-specific data storage. By combining these modern cybersecurity techniques into a functional system, this study demonstrates how secure design principles can be applied in real-world applications. It also serves as an educational model for understanding encryption workflows, authentication mechanisms, and secure backend architecture. The case study contributes both academically and practically by presenting a scalable, transparent, and implementable approach to secure password management aligned with contemporary cybersecurity standards.

5 Methodology

5.1 Approach of the Course-End Project

The course-end project was carried out by designing and developing a secure web-based password management system. The project focused on implementing modern cybersecurity techniques such as encryption, hashing, and token-based authentication. A full-stack development approach was followed, integrating frontend, backend, and database components to build a functional and secure system.

5.2 Data Collection Methods

The project was implemented and evaluated through:

1. Simulation and testing of user registration, login, and password storage processes.
2. System logs and API responses to verify authentication and encryption workflows.
3. Functional testing to ensure proper encryption, decryption, and user-specific data separation.

The project primarily focused on system development and testing rather than survey-based data collection.

5.3 Assumptions

Users create strong login and master passwords.

The server environment is properly secured.

JWT secret keys and database credentials are safely stored.

The system is designed for web-based usage only.

5.4 System Architecture / Workflow

The system follows a client-server architecture:

1. User registers → Passwords are hashed using bcrypt.
2. User logs in → A JWT token is generated for authentication.
3. User verifies master password → Access to vault is granted.
4. Vault passwords are encrypted using AES before storing in the database.
5. Passwords are decrypted only after successful authentication.

This workflow ensures secure authentication, encrypted storage, and user-specific access control.

6 System Description

The system is developed in a web-based environment where users need a secure method to store and manage multiple online account credentials. The environment consists of a frontend interface built using React.js, a backend server developed using Node.js and Express.js, and a cloud-hosted MongoDB Atlas database for data storage.

Users interact with the system through a browser interface. The backend handles authentication, encryption, and database operations. Sensitive information such as passwords is never stored in plain text. Instead, login passwords and master passwords are hashed using bcrypt, while stored credentials are encrypted using AES-256 encryption before being saved in the database. JWT-based authentication ensures secure session management.

The system operates in a secure client-server architecture where all sensitive operations are handled on the server side.

6.1 Flow of Events

The system workflow is as follows:

User Registration

1. User enters name, email, password, and master password.
2. Password and master password are hashed using bcrypt.
3. User data is stored securely in the database.

User Login

1. User enters email and password.
2. System verifies credentials.
3. JWT token is generated for session authentication.

Master Password Verification

1. User enters master password.
2. System verifies it using bcrypt.
3. Access to password vault is granted.

Add Password

1. User enters website, username, and password.
2. Password is encrypted using AES-256.
3. Encrypted data is stored in the database with user ID.

View Password

1. System retrieves encrypted passwords.
2. Decryption occurs only after successful authentication.

Logout

1. JWT token is removed from local storage.
2. Session ends securely.

7 Results



Fig 1: Access Page: This page acts as the entry point of the SecureVault system. It provides options for users to either log in or register to access the password management features.

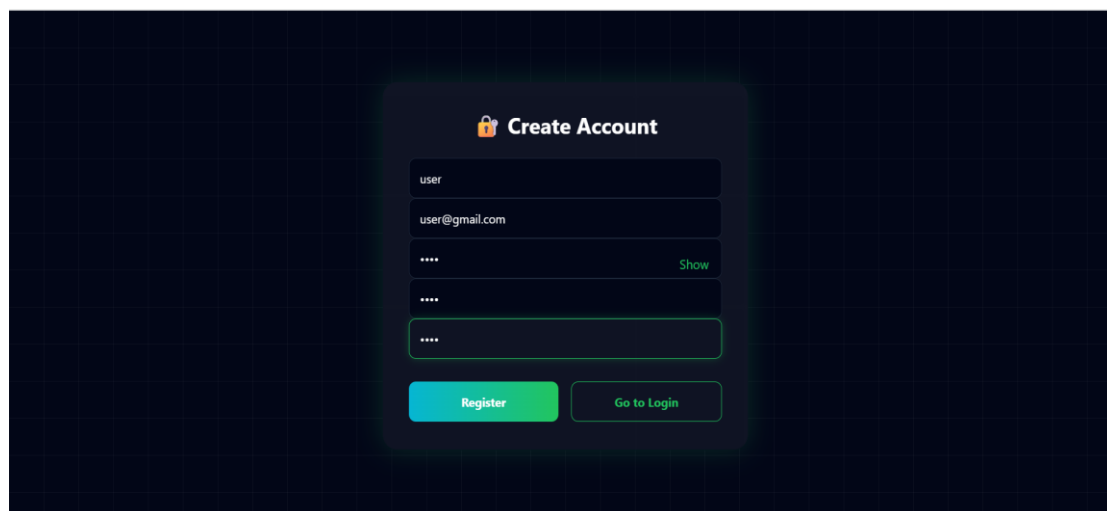


Fig 2: Register page: This page acts as the entry point of the SecureVault system. It provides options for users to either log in or register to access the password management features.

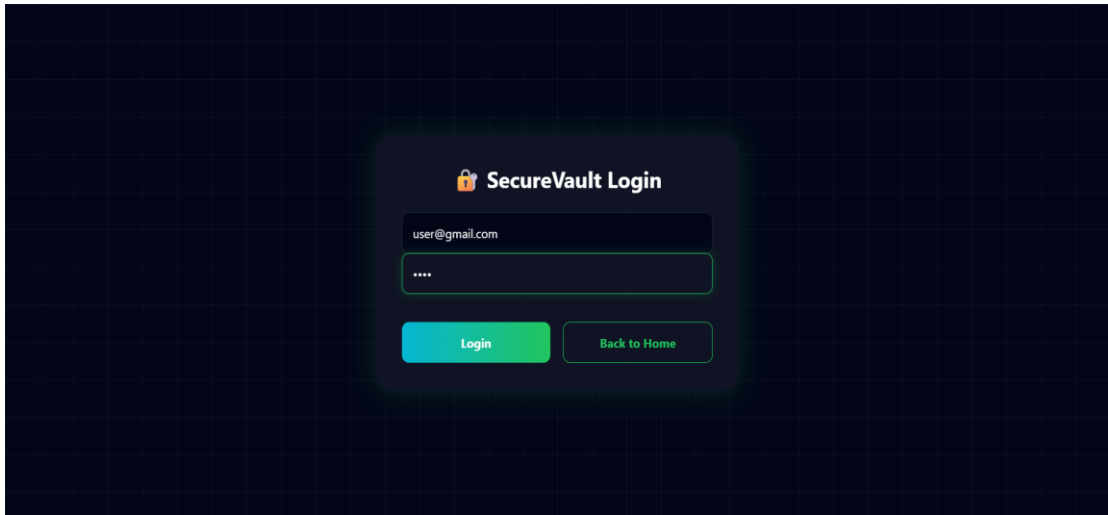


Fig 3: Login page: The Register page allows new users to create an account by entering their personal details and passwords. The credentials are securely processed and stored in the system.

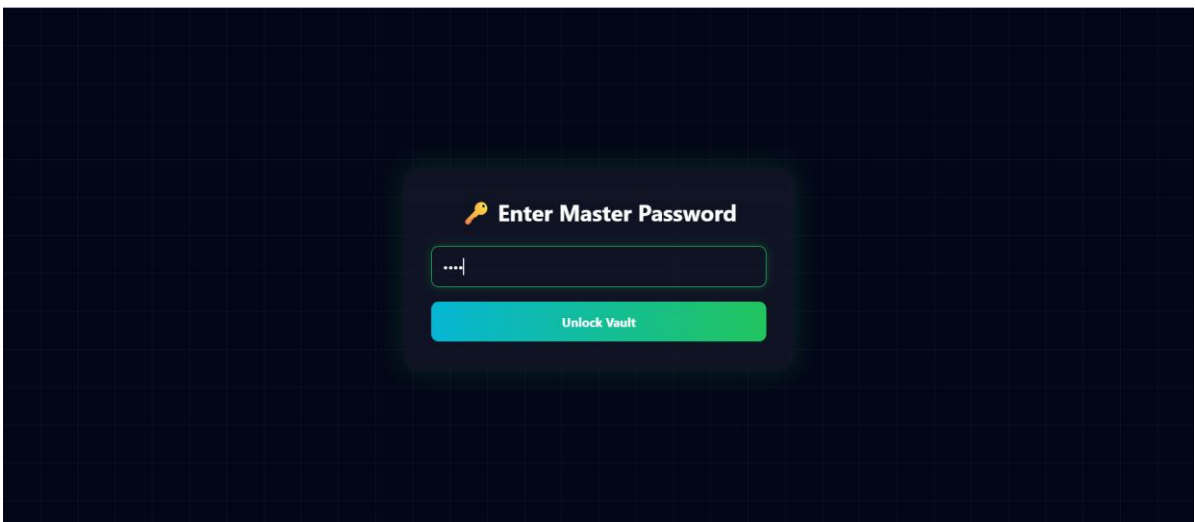


Fig 4: Master Key: The Login page enables registered users to access the system using their email and password. The entered credentials are verified before granting access.

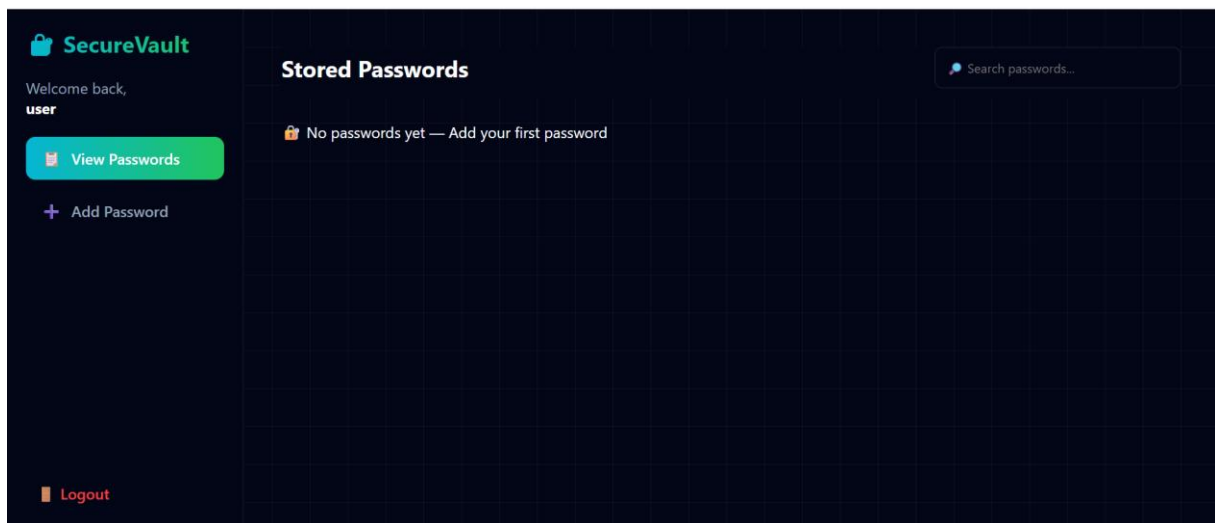


Fig 5: Dashboard page: The Dashboard is the main interface where users can manage their stored credentials. It provides options to view saved passwords and add new ones.

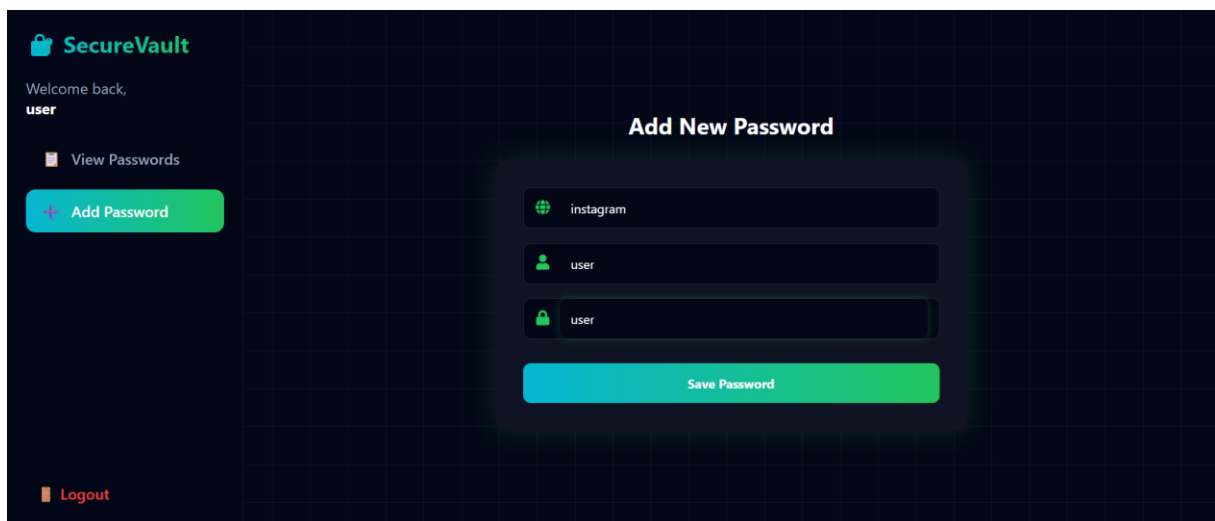


Fig 6: Add password: This page allows users to store new login credentials such as website name, username, and password. The password is encrypted before being stored in the database.

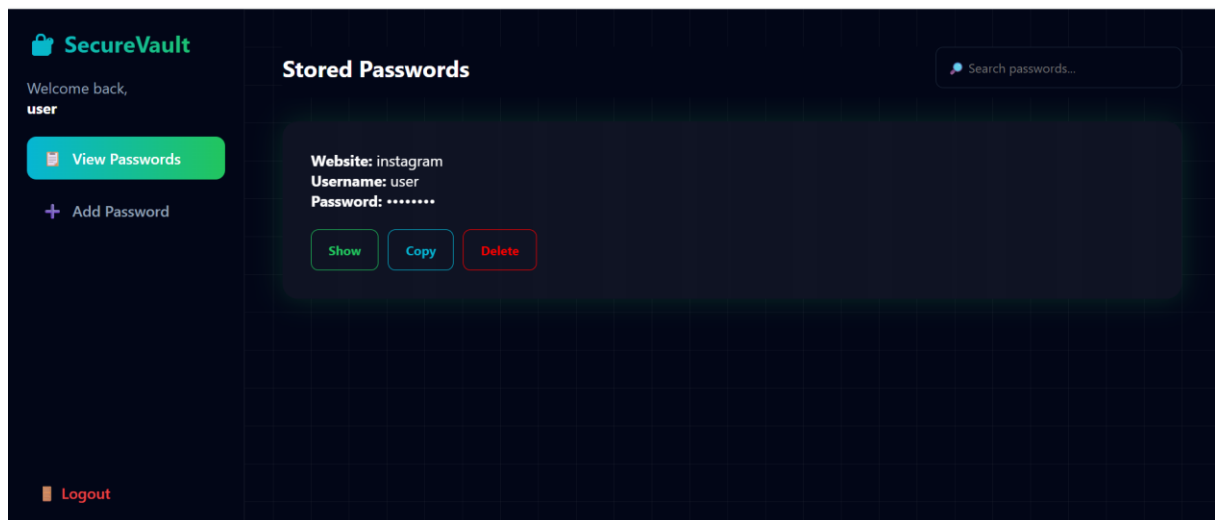


Fig 7: View password: This page allows users to store new login credentials such as website name, username, and password. The password is encrypted before being stored in the database.

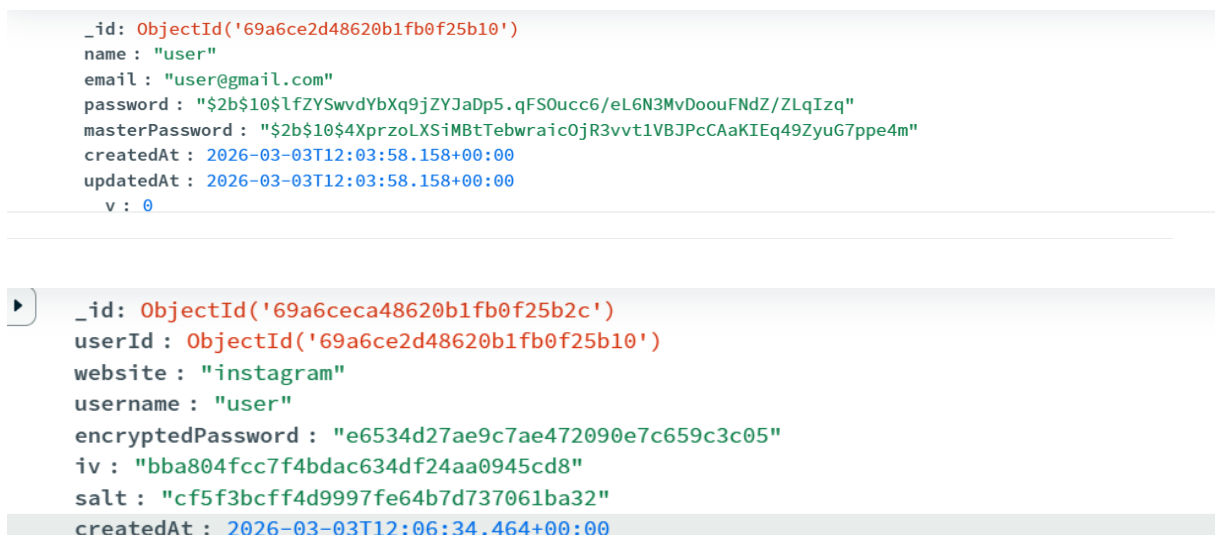


Fig 8: Encrypted data: The View Password page displays the list of stored credentials for the user. It allows users to manage their passwords with options like show, copy, and delete.

8. Conclusion

The Secure Password Management System developed in this course-end project successfully demonstrates the implementation of modern cybersecurity techniques in a real-world web application. The system ensures secure storage of user credentials by applying AES-256 encryption for vault passwords and bcrypt hashing for login and master passwords. JWT-based authentication provides secure session management and controlled access to user data.

The project effectively addresses the problem of insecure password storage and password reuse by providing a structured, encrypted, and user-specific solution. Each user's data is isolated in the database, ensuring privacy and protection against unauthorized access. The integration of frontend, backend, and database technologies demonstrates a complete full-stack security-oriented architecture.

Overall, the project highlights the importance of encryption, secure authentication, and proper data handling in modern web applications. It serves as a practical example of applying cybersecurity principles to build a secure and scalable password management system.

9. Future Scope

Although the Secure Password Management System provides secure storage and authentication features, several improvements can be implemented in the future to enhance functionality and security.

1. Multi-FactorAuthentication(MFA):

Adding OTP or biometric-based authentication can improve account security beyond single-password verification.

2. PasswordStrengthAnalyzer:

Integrating a real-time password strength checker can help users create stronger passwords.

3. AutoPasswordGenerator:

A built-in strong password generator can assist users in generating secure and complex passwords.

4. MobileApplicationSupport:

Developing Android or iOS versions of the system would increase accessibility and usability.

5. CloudBackupandSync:

Enabling encrypted cloud backup and multi-device synchronization can improve convenience.

10. References

- [1] W. Stallings, *Cryptography and Network Security: Principles and Practice*, 7th ed., Pearson Education, 2017.
- [2] National Institute of Standards and Technology (NIST), “Digital Identity Guidelines,” NIST Special Publication 800-63B, 2017.
- [3] Open Web Application Security Project (OWASP), “OWASP Top 10: The Ten Most Critical Web Application Security Risks,” 2021.
- [4] A. Biryukov, D. Dinu, and D. Khovratovich, “Argon2: The Memory-Hard Function for Password Hashing and Other Applications,” IEEE European Symposium on Security and Privacy, 2016.
- [5] N. Provos and D. Mazières, “A Future-Adaptable Password Scheme,” USENIX Annual Technical Conference, 1999.
- [6] M. Jones, J. Bradley, and N. Sakimura, “JSON Web Token (JWT),” IETF RFC 7519, 2015.
- [7] MongoDB Inc., “MongoDB Documentation,” Available: <https://www.mongodb.com/docs>
- [8] Node.js Foundation, “Node.js Documentation,” Available: <https://nodejs.org>